

1).

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem1
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem1> node server
Server is listening on port number : 8000
```



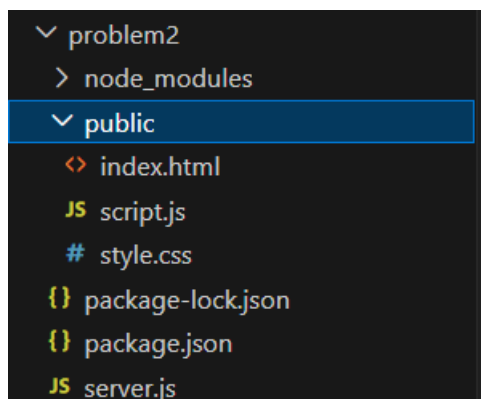
NodeJS AJAX Example

Get Hello

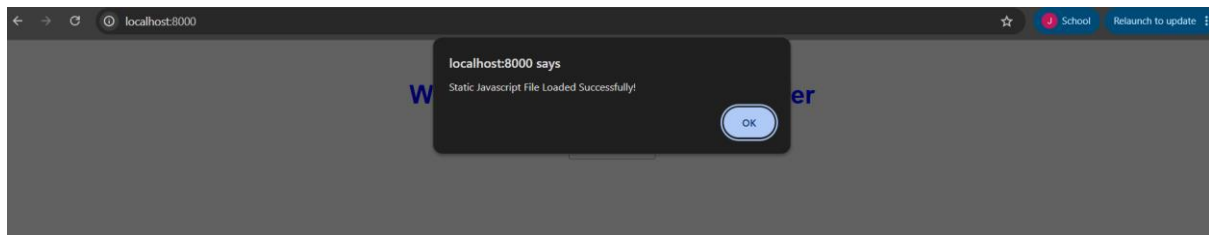
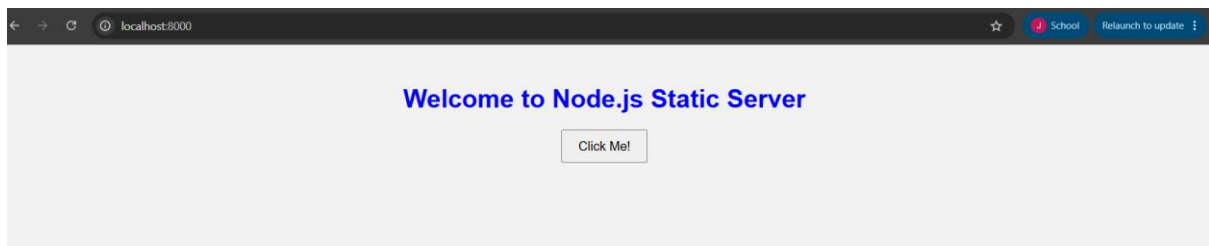


Hello NodeJS!!

2).



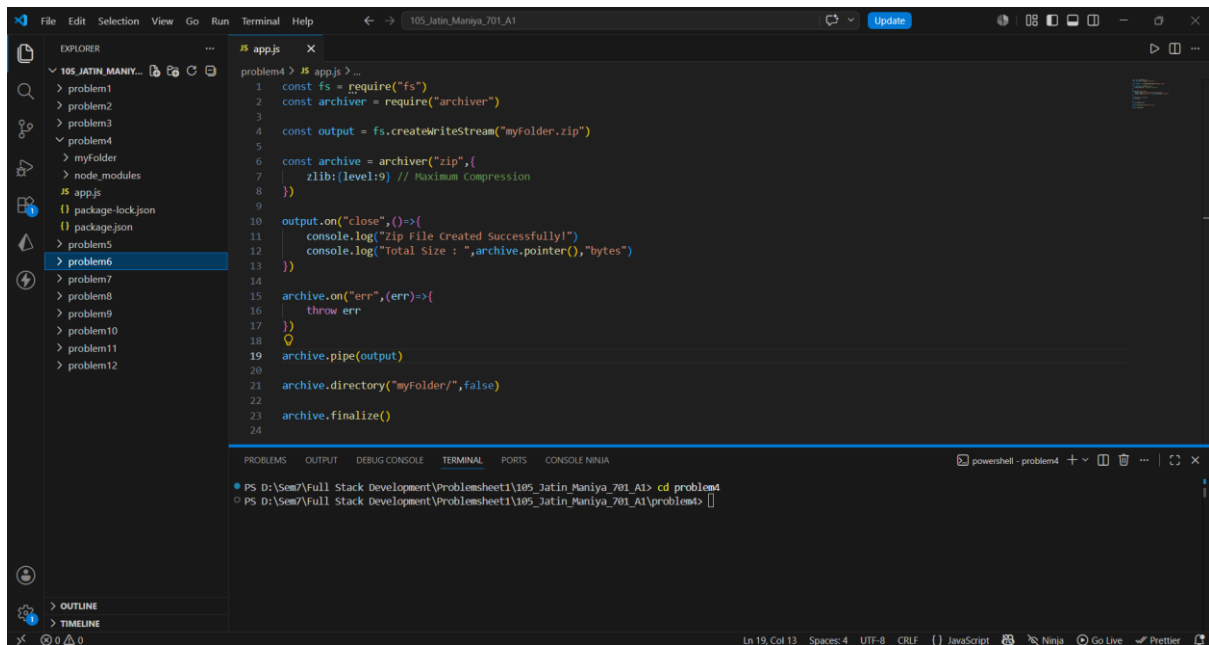
```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem2
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem2> node server
Server is listening on port number 8000
```



3).

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem3> node app
===== College Chatbot =====
Type 'bye' to exit.
You : hi
Bot : Hello! Welcome to ABC College.
You : What courses are available?
Bot : We offer BCA, B.Sc IT, B.Com and MBA.
You : When are admissions open?
Bot : Admissions are open from June to August.
You : Where is the college located?
Bot : ABC College is located on College Road, Surat, Gujarat.
You : Is hostel available?
Bot : Yes, hostel facilities are available for both boys and girls.
You : thank you
Bot : Thank you. Have a nice day!
```

4).

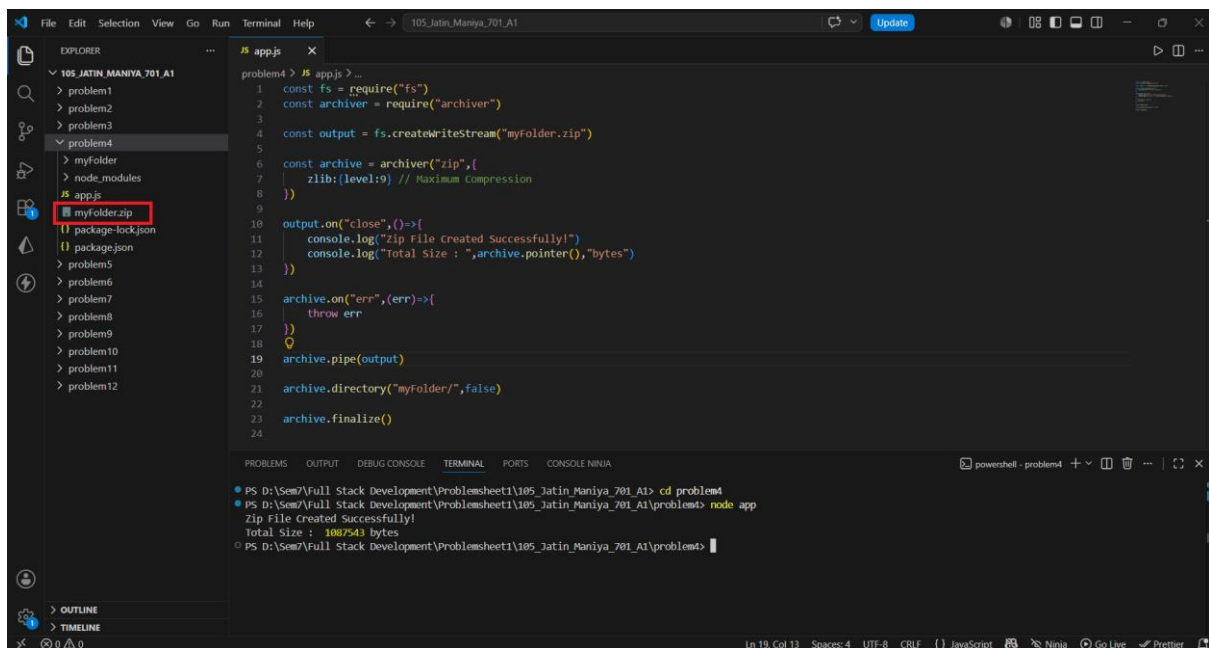


The image shows a Visual Studio Code editor window. The Explorer panel on the left shows a project structure with a folder named '105_JATIN_MANIYA_701_A1' containing subfolders 'problem1' through 'problem12'. The 'app.js' file is selected in the Explorer. The main editor area displays the code for 'app.js', which is a JavaScript script using the 'archiver' and 'fs' modules to create a zip file named 'myfolder.zip'. The code includes error handling and logging. The Terminal panel at the bottom shows the command prompt with the current directory set to 'D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem4'.

```
1 const fs = require("fs")
2 const archiver = require("archiver")
3
4 const output = fs.createWriteStream("myfolder.zip")
5
6 const archive = archiver("zip",{
7   zlib:{level:9} // Maximum Compression
8 })
9
10 output.on("close",()=>{
11   console.log("Zip File Created Successfully!")
12   console.log("Total Size : ",archive.pointer(),"bytes")
13 })
14
15 archive.on("err",(err)=>{
16   throw err
17 })
18
19 archive.pipe(output)
20
21 archive.directory("myfolder/",false)
22
23 archive.finalize()
24
```

Terminal output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem4
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem4>
```



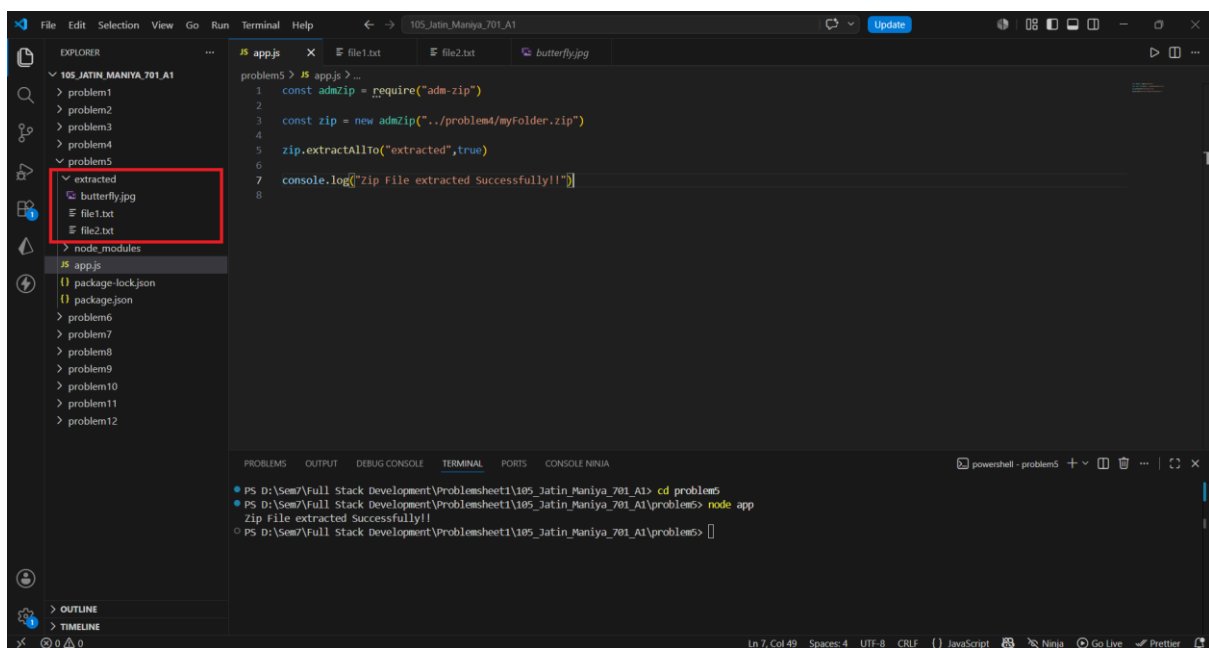
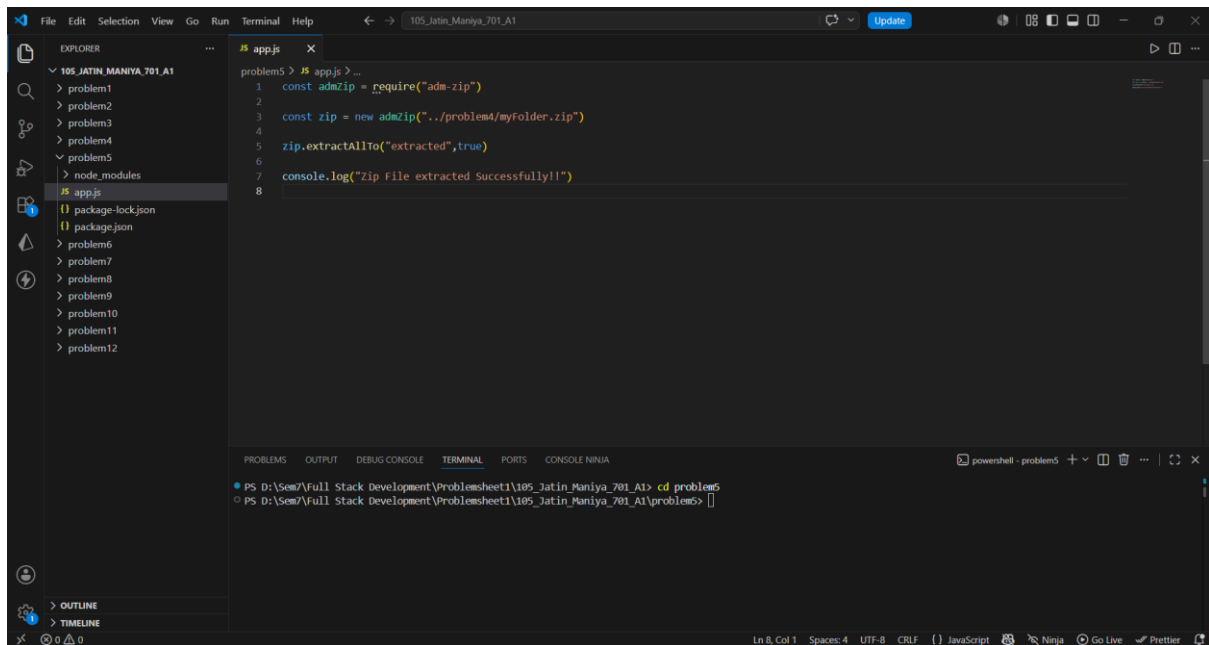
The image shows the same Visual Studio Code editor window as before, but now the 'myfolder.zip' file is visible in the Explorer panel under the 'problem4' folder. The main editor area still shows the 'app.js' code. The Terminal panel at the bottom shows the command prompt with the current directory set to 'D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem4'. The terminal output shows the successful execution of the 'node app' command, indicating that the zip file was created successfully with a total size of 1087543 bytes.

```
1 const fs = require("fs")
2 const archiver = require("archiver")
3
4 const output = fs.createWriteStream("myfolder.zip")
5
6 const archive = archiver("zip",{
7   zlib:{level:9} // Maximum Compression
8 })
9
10 output.on("close",()=>{
11   console.log("Zip File Created Successfully!")
12   console.log("Total Size : ",archive.pointer(),"bytes")
13 })
14
15 archive.on("err",(err)=>{
16   throw err
17 })
18
19 archive.pipe(output)
20
21 archive.directory("myfolder/",false)
22
23 archive.finalize()
24
```

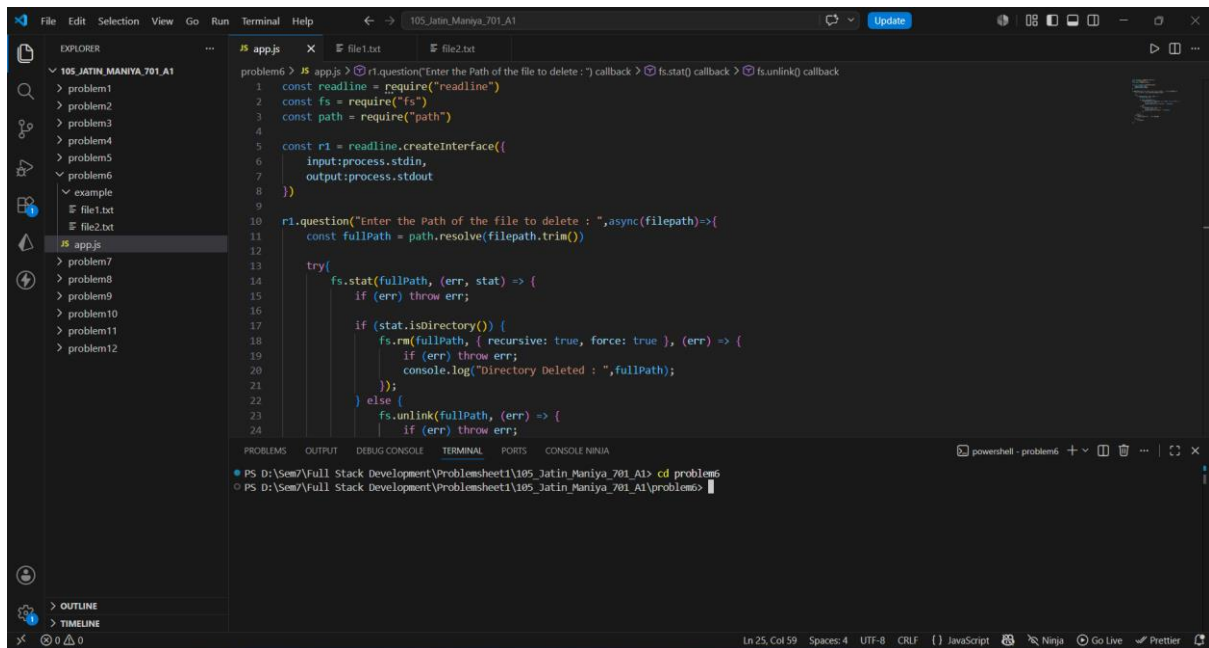
Terminal output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem4
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem4> node app
Zip File Created Successfully!
Total Size : 1087543 bytes
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem4>
```

5).



6).

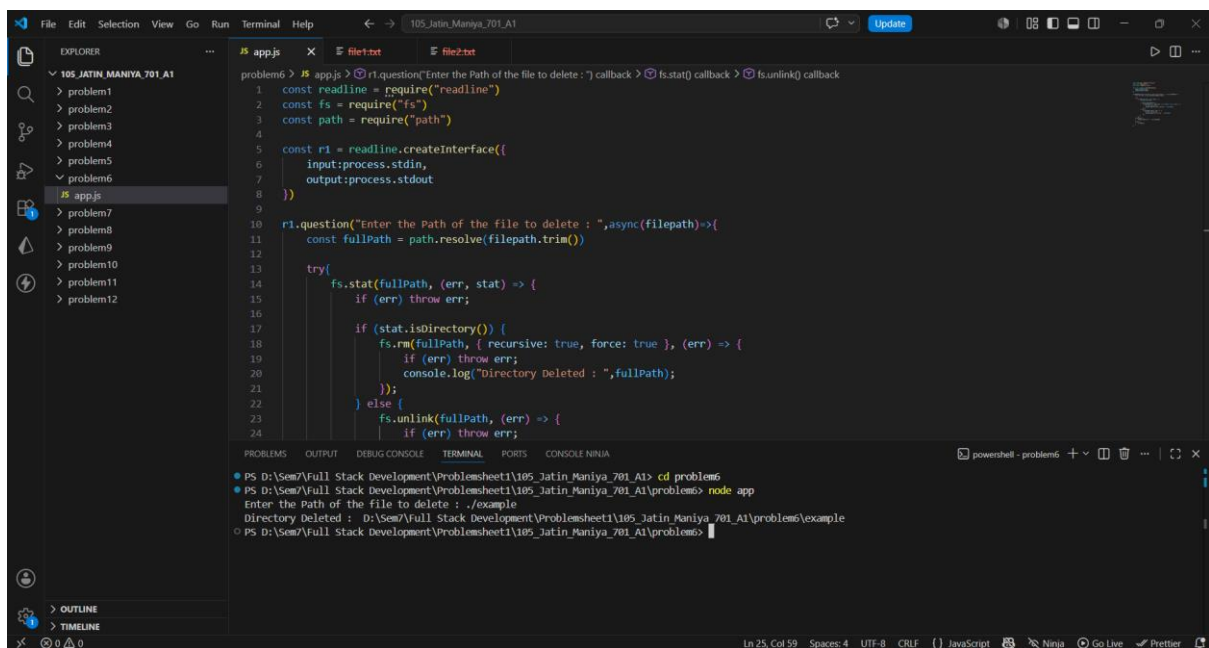


The image shows a VS Code editor window with the file explorer on the left, the editor in the center, and the terminal at the bottom. The file explorer shows a project named '105_JATIN_MANIYA_701_A1' with a folder 'example' containing 'file1.txt' and 'file2.txt'. The editor shows the 'app.js' file with the following code:

```
1 const readline = require("readline")
2 const fs = require("fs")
3 const path = require("path")
4
5 const rl = readline.createInterface({
6   input: process.stdin,
7   output: process.stdout
8 })
9
10 rl.question("Enter the Path of the file to delete : ", async (filepath) => {
11   const fullPath = path.resolve(filepath.trim())
12
13   try {
14     fs.stat(fullPath, (err, stat) => {
15       if (err) throw err;
16
17       if (stat.isDirectory()) {
18         fs.rm(fullPath, { recursive: true, force: true }, (err) => {
19           if (err) throw err;
20           console.log("Directory Deleted : ", fullPath);
21         });
22       } else {
23         fs.unlink(fullPath, (err) => {
24           if (err) throw err;
25         });
26       }
27     });
28   } catch (err) {
29     console.log("Error: ", err);
30   }
31 })
```

The terminal shows the following output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problems
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problems>
```



The image shows a VS Code editor window with the file explorer on the left, the editor in the center, and the terminal at the bottom. The file explorer shows a project named '105_JATIN_MANIYA_701_A1' with a folder 'example' containing 'file1.txt' and 'file2.txt'. The editor shows the 'app.js' file with the following code:

```
1 const readline = require("readline")
2 const fs = require("fs")
3 const path = require("path")
4
5 const rl = readline.createInterface({
6   input: process.stdin,
7   output: process.stdout
8 })
9
10 rl.question("Enter the Path of the file to delete : ", async (filepath) => {
11   const fullPath = path.resolve(filepath.trim())
12
13   try {
14     fs.stat(fullPath, (err, stat) => {
15       if (err) throw err;
16
17       if (stat.isDirectory()) {
18         fs.rm(fullPath, { recursive: true, force: true }, (err) => {
19           if (err) throw err;
20           console.log("Directory Deleted : ", fullPath);
21         });
22       } else {
23         fs.unlink(fullPath, (err) => {
24           if (err) throw err;
25         });
26       }
27     });
28   } catch (err) {
29     console.log("Error: ", err);
30   }
31 })
```

The terminal shows the following output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problems
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problems> node app
Enter the path of the file to delete : ./example
Directory Deleted : D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problems\example
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problems>
```

[illegible]

The screenshot shows the VS Code interface with the Explorer, Explorer, and Terminal panels. The Explorer panel on the left shows a file tree for '105_Jatin_Maniya_701_A1' with files like 'problem1', 'problem2', ..., 'package.json', 'systemInfo.js', 'problem9', 'problem10', 'problem11', and 'problem12'. The Explorer panel in the center shows the content of 'package.json' with the following JSON:

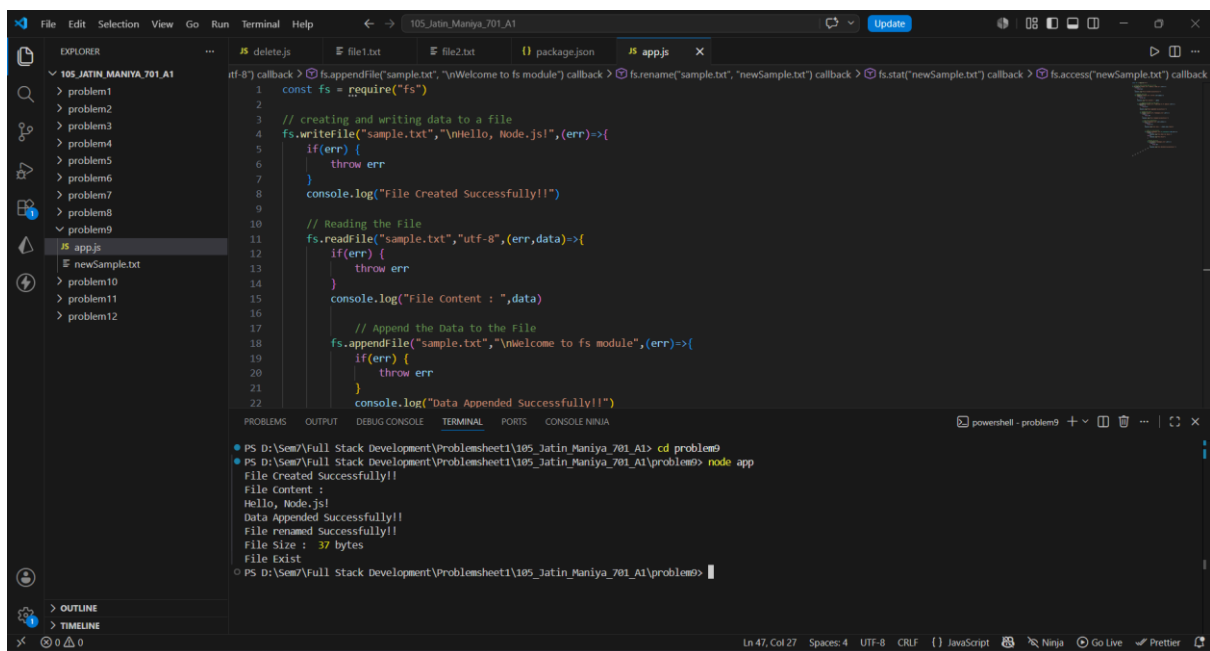
```
{
  "name": "problem8",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "start": "node index.js",
    "test": "echo \"Error: Running test cases...\" && exit 1",
    "dev": "nodemon index.js",
    "systemInfo": "node systemInfo.js",
    "delete-folder": "node delete.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

The Terminal panel at the bottom shows the output of the command 'npm run systeminfo' executed in the directory 'PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem8'. The output indicates the current platform is win32, the time is 7/15/2026, 9:42:34 AM, and the command 'node systemInfo.js' was executed. The output of the script is as follows:

```
problem8@1.0.0 dev
nodemon index.js

[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting 'node index.js'
Server is Listening on port number : 8000
```

9).

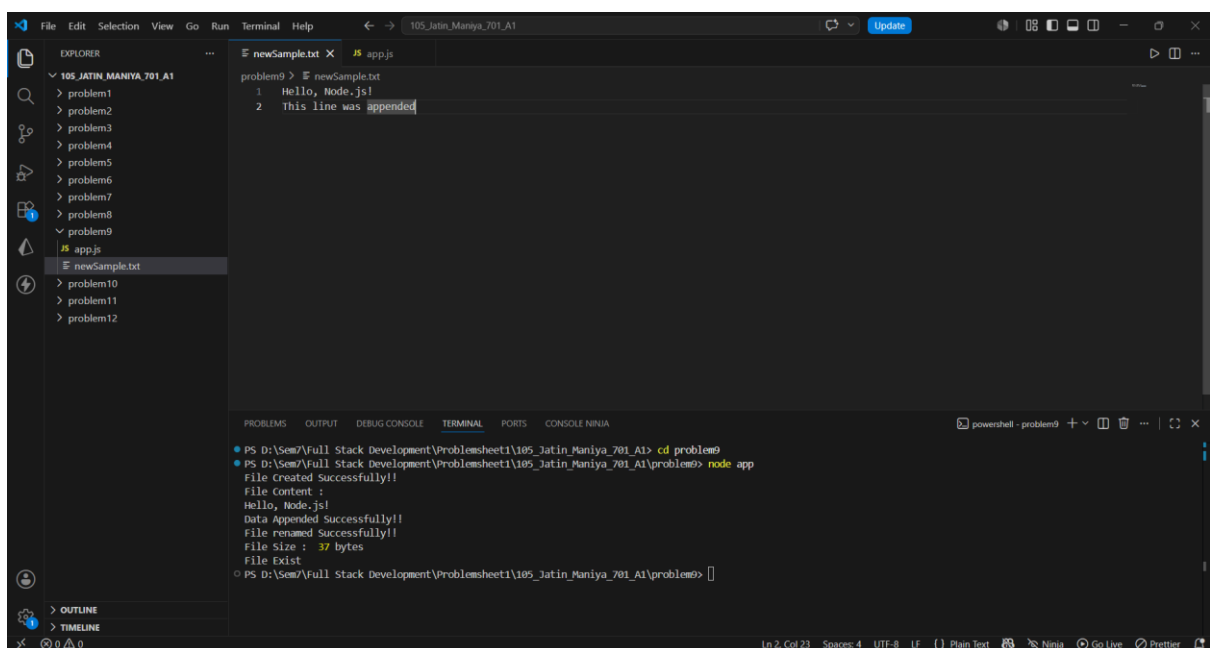


The screenshot shows the VS Code editor with the file explorer on the left displaying a project named '105_JATIN_MANIYA_701_A1'. The file 'app.js' is selected. The editor window shows the following JavaScript code:

```
1  const fs = require("fs")
2
3  // creating and writing data to a file
4  fs.writeFile("sample.txt", "Hello, Node.js!", (err) => {
5    if (err) {
6      throw err
7    }
8    console.log("File Created Successfully!!")
9  })
10
11 // Reading the File
12 fs.readFile("sample.txt", "utf-8", (err, data) => {
13   if (err) {
14     throw err
15   }
16   console.log("File Content : ", data)
17 }
18
19 // Append the Data to the File
20 fs.appendFile("sample.txt", "Hello, Node.js!", (err) => {
21   if (err) {
22     throw err
23   }
24   console.log("Data Appended Successfully!!")
25 })
```

The terminal at the bottom shows the following output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem9
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem9> node app
File Created Successfully!!
File Content :
Hello, Node.js!
Data Appended Successfully!!
File renamed Successfully!!
File Size : 37 bytes
File Exist
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem9>
```



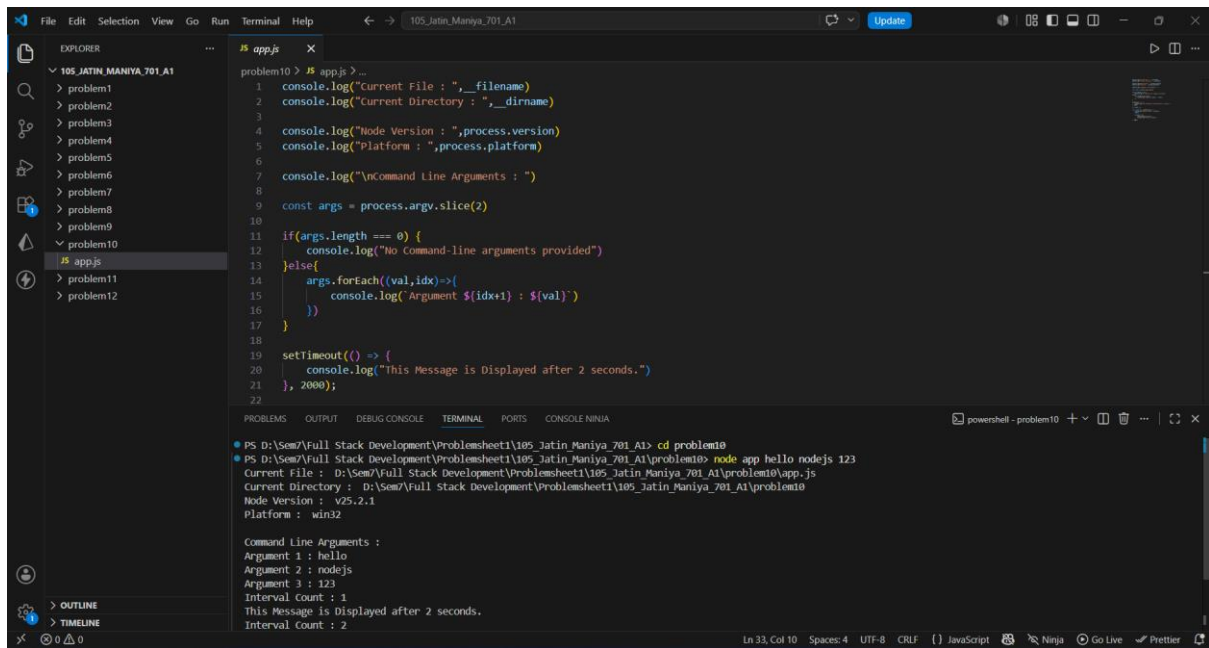
The screenshot shows the VS Code editor with the file explorer on the left displaying the same project. The file 'newSample.txt' is selected. The editor window shows the following text:

```
1 Hello, Node.js!
2 This line was appended
```

The terminal at the bottom shows the following output:

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem9
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem9> node app
File Created Successfully!!
File Content :
Hello, Node.js!
Data Appended Successfully!!
File renamed Successfully!!
File Size : 37 bytes
File Exist
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem9>
```


10).



The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left displays a project named '105_JATIN_MANIYA_701_A1' with a list of files including 'problem1' through 'problem12', and 'app.js' which is currently selected. The main editor window shows the content of 'app.js', a JavaScript script that logs file information, Node.js version, platform, and command-line arguments, and includes a 2-second timeout. The integrated terminal at the bottom shows the command 'node app hello nodejs 123' being executed, with its output displayed below the command line.

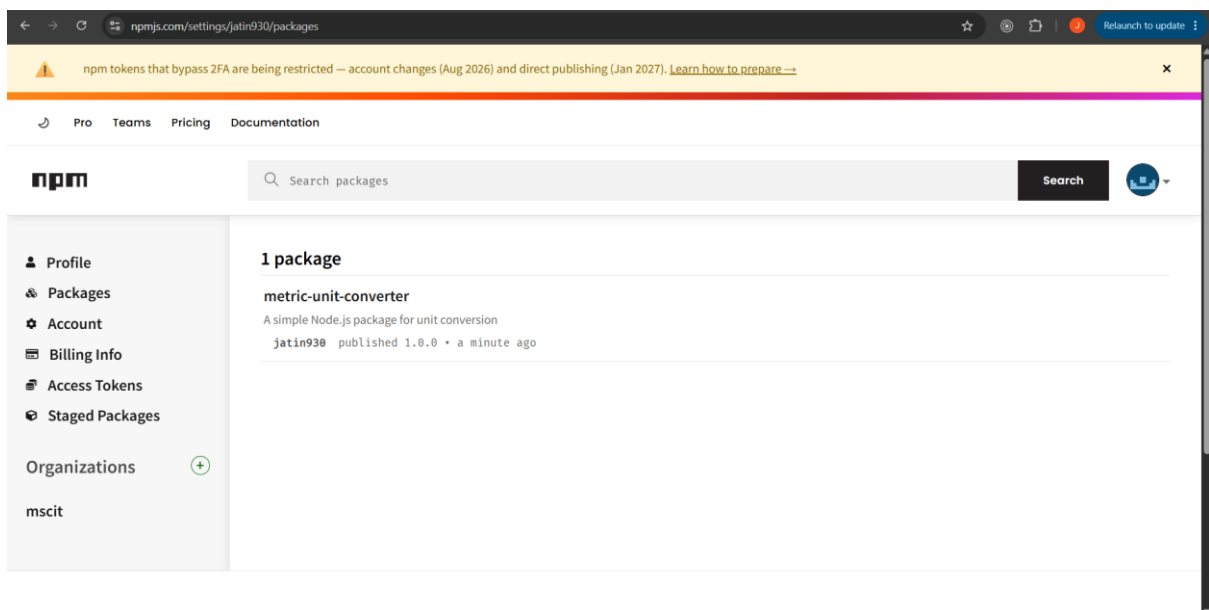
```
problem10 > .\app.js >...
1 console.log("Current File : ", __filename)
2 console.log("Current Directory : ", __dirname)
3
4 console.log("Node Version : ", process.version)
5 console.log("Platform : ", process.platform)
6
7 console.log("\ncommand Line Arguments : ")
8
9 const args = process.argv.slice(2)
10
11 if(args.length === 0) {
12   console.log("No Command-line arguments provided")
13 }else{
14   args.forEach((val,idx)=>{
15     console.log("Argument ${idx+1} : ${val}")
16   })
17 }
18
19 setTimeout(() => {
20   console.log("This Message is Displayed after 2 seconds.")
21 }, 2000);
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS CONSOLE NINJA

```
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1> cd problem10
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem10> node app hello nodejs 123
Current File : D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem10\app.js
Current Directory : D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem10
Node Version : v25.2.1
Platform : win32

Command Line Arguments :
Argument 1 : hello
Argument 2 : nodejs
Argument 3 : 123
Interval Count : 1
This Message is Displayed after 2 seconds.
Interval Count : 2
```

11).




```
package.json  test.js  app.js

problem11 > metric-unit-converter > test.js > -
1  const converter = require("metric-unit-converter")
2
3  console.log("10 km = ",converter.kmToMiles(10),"miles")
4  console.log("6.2 miles = ",converter.milesToKm(6.2),"km")
5
6  console.log("5 kg = ",converter.kgToPounds(5),"pounds")
7  console.log("11 pounds = ",converter.poundsToKg(11),"kg")
8
9  console.log("100 cm = ",converter.cmToInches(100),"inches")
10 console.log("39 inches = ",converter.inchesToCm(39),"cm")
11
12 console.log("25°C = ",converter.celsiusToFahrenheit(25),"°F")
13 console.log("77°F = ",converter.fahrenheitToCelsius(77),"°C")
14

PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem11\metric-unit-converter> npm publish
Press ENTER to open in the browser...

+ metric-unit-converter@1.0.0
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem11\metric-unit-converter> npm i metric-unit-converter
added 1 package, and audited 2 packages in 3s

found 0 vulnerabilities
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem11\metric-unit-converter> node test
10 km = 6.21 miles
6.2 miles = 9.98 km
5 kg = 11.02 pounds
11 pounds = 4.99 kg
100 cm = 39.37 inches
39 inches = 99.06 cm
25°C = 77.00 °F
77°F = 25.00 °C
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem11\metric-unit-converter> |
```

12).

```
problem12 > app.js >
1  function addtoCart() {
2    return new Promise((resolve)=>{
3      setTimeout(() => {
4        console.log("Product Added to Cart")
5        resolve()
6      }, 1000);
7    })
8  }
9
10 function makePayment() {
11   return new Promise((resolve)=>{
12     setTimeout(() => {
13       console.log("Payment Successful")
14       resolve()
15     }, 1000);
16   })
17 }
18
19 function placeOrder() {
20   return new Promise((resolve)=>{
21     setTimeout(() => {
22       console.log("Order Placed Successfully")
23       resolve()
24     }, 1000);
25   })
26 }
27
28 addtoCart()

PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem12> node app
Product Added to Cart
Payment Successful
Order Placed Successfully
Shopping Completed
PS D:\Sem7\Full Stack Development\Problemsheet1\105_Jatin_Maniya_701_A1\problem12> |
```